Short 3

Metric Spaces

3.1 Metric

Let X be a nonempty set.

3.1.1 DEFINITION. *A function $\rho : X \times X \to \mathbb{R}$ satisfying the following properties:*

$(\mathcal{M}_1)$ $\quad \rho(x, y) = 0 \Leftrightarrow x = y$;

$(\mathcal{M}_2)$ $\quad \rho(x, y) \leq \rho(x, z) + \rho(y, z)$ $\quad$ *(the triangle inequality);*

for all $x, y, z \in X$, is called a metric *on X.*

3.1.2 DEFINITION. *The set X along with a metric ρ is called a* metric space *and is denoted by (X, ρ).*

The elements of (X, ρ) are called points and the number $\rho(x, y)$ is called the distance between the points x and y.

3.1.3 THEOREM. *A metric $\rho : X \times X \to \mathbb{R}$ has the following properties:*

(M_1) $\quad \rho(x, y) \geq 0$ *and* $\rho(x, y) = 0 \Leftrightarrow x = y$;

(M_2) $\quad \rho(x, y) = \rho(y, x)$, *(symmetry);*

(M_3) $\quad \rho(x, y) \leq \rho(x, z) + \rho(z, y)$, *(the triangle inequality);*

for all $x, y, z \in X$.

3.1.4 EXAMPLE. The function $\rho_1 : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$,

$$\rho_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|,$$

where $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$, is a metric called Minkovski metric on $\mathbb{R}^n$.

3.1.5 EXAMPLE. The function $\rho_2 : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$,

$$\rho_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2},$$

where $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$, is a metric called Euclidean metric on $\mathbb{R}^n$.

3.1.6 EXAMPLE. The function $\rho_\infty : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$,

$$\rho_\infty(x, y) = \max_{i=1,\dots,n} \{|x_i - y_i|\},$$

where $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, is a metric called the *Chebyshev metric* on $\mathbb{R}^n$.

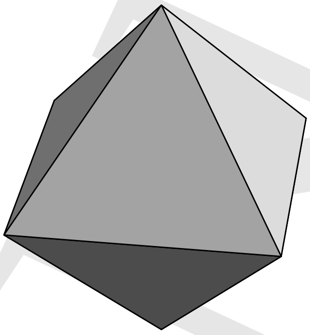
Let (X, ρ) be a metric space, $\emptyset \neq A, B \subseteq X$ and $r > 0$. The following notions are frequently used:

- the *open ball centered at x of radius r*

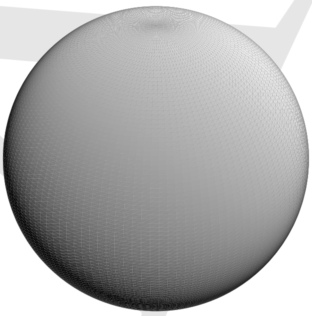
$$B(x, r) = \{y \mid (y \in X) \wedge (\rho(x, y) < r)\};$$

- the *closed ball centered at x of radius r*

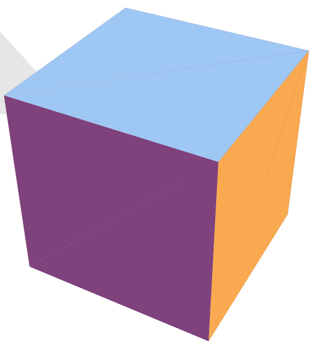
$$\overline{B}\,(x, r) = \{y \mid (y \in X) \wedge (\rho(x, y) \leq r)\};$$



The Minkovski ball $B((0,0,0),1)$
$= \left\{(x,y,z) \in R^3 \,\middle|\, |x| + |y| + |z| \leq 1\right\}$



The Euclidean ball $B((0,0,0),1)$
$= \left\{(x,y,z) \in R^3 \,\middle|\, \sqrt{x^2 + y^2 + z^2} \leq 1\right\}$



The Chebyshev ball $B((0,0,0),1)$
$= \left\{(x,y,z) \in R^3 \,\middle|\, \max\{|x|, |y|, |z|\} \leq 1\right\}$

- the **distance between the point x and the set A**

$$\mathrm{d}(x, A) = \inf_{a \in A} \rho(x, a);$$

- the **distance between the set A and the set B**

$$\mathrm{d}(A, B) = \inf_{a \in A,\, b \in B} \rho(a, b).$$

3.2 Topology of a Metric Space

3.2.1 THEOREM. *In a metric space (X, ρ) we can define the following topology*

$$\mathcal{T} = \{\emptyset, G \mid (G \subseteq X) \wedge (\forall x)(\exists r)((x \in G) \to (r > 0 \wedge B(x, r) \subseteq G))\}.$$

Proof.

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3.2.2 REMARK. *In the topology induced by the metric ρ the open balls are open sets.*

3.2.3 REMARK. *Unless differently specified, by the topology of a metric space we mean the topology induced by the metric of the space.*

3.3 Sequences in Metric Spaces

Let (X, ρ) be a metric space. We consider the set

$$X^{\mathbb{N}} = X \times X \times \cdots \times X \times \cdots = \{f \mid f : \mathbb{N} \to X\}.$$

3.3.1 DEFINITION. *An element f in the set $X^{\mathbb{N}}$ is called a* **sequence**.

Using the notation $f(n) = x_n$, we shall denote the sequence $(x_0, x_1, \ldots, x_n, \ldots)$ by the symbol

$$(x_n).$$

The element x_n is called the n^{th} **term** of the sequence.

3.3.2 DEFINITION. *The sequence (x_n) is said to be* **convergent** *if there exists a point $x \in X$ such that, for all $\varepsilon > 0$, there exists a finite number of terms x_n lying outside the ball $B(x, \varepsilon)$.*

The point x is called the **limit of the sequence (x_n)**. The following notations will be used:

$$\lim_{n \to \infty} x_n = x \quad \text{or} \quad x_n \to x.$$

3.3.3 DEFINITION. *A sequence (x_n) in the metric space (X, ρ) is called a* **fundamental sequence** *or a* **Cauchy sequence** *iff, given any $\varepsilon > 0$ there exists an r_ε such that*

$$\rho(x_n, x_m) < \varepsilon,$$

provided $n, m \in \mathbb{N},\ n, m > r_\varepsilon$.

3.3.4 DEFINITION. *A metric space* (X, ρ) *is said to be* complete *iff any Cauchy sequence in* X *converges to a point of* X.

3.3.5 EXAMPLE. The space $\mathbb{Q}$ along with the Euclidean metric

$$\rho(x, y) = |x - y|$$

is not complete space. Although the sequence

$$\left(\left(1 + \frac{1}{n}\right)^n\right) \in \mathbb{Q}^{\mathbb{N}^*}$$

is a Cauchy sequence, however

$$\left(1 + \frac{1}{n}\right)^n \to \mathsf{e} \notin \mathbb{Q}.$$

3.3.6 REMARK. *Every convergent sequence is a Cauchy sequence.*

3.3.7 REMARK. *In order to prove that a sequence* (x_n) *is a Cauchy sequence it is enough to find a sequence of positive real numbers* (ε_n), *which converges to zero, such that*

$$\rho(x_{n+p}, x_n) \leq \varepsilon_n,$$

provided $n, p \in \mathbb{N}$.

3.4 Bounded Sets in Metric Spaces

3.4.1 DEFINITION. *A set* A *in a metric space* (X, ρ) *is said to be* bounded *iff there exists a point* $x \in X$ *and a number* $r > 0$ *such that*

$$A \subseteq B(x, r).$$

3.4.2 THEOREM. *Any compact set in a metric space is bounded.*

Proof. Let (X, ρ) be metric space, let $A \subseteq X$ be a compact set and let $x \in X$. Since the family

$$\{B(x, n)\}_{n \in \mathbb{N}}$$

is a open covering of X and, implicitly, an open cover of the set A and A is compact, then there exists an finite open subcover

$$\{B(x, n_i)\}_{i = \overline{1, m}}$$

of the set A, where $n_1 < n_2 < \ldots < n_m$. We have

$$A \subseteq \bigcup_{i=1}^{m} B(x, n_i) = B(x, n_m),$$

i.e. the set A is bounded. ◇◇◇